DO NOW — RECALL

Timed Paper 1 data and logic practice

Name: _____ Class: _____ Date: _____

1. Which sort builds an ordered prefix by inserting a key?

2. How do you compensate for shifting a positive fractional mantissa left?

3. Simplify ¬(A ∨ B).

4. Find the decimal value of $0.1100_2 \times 2^3$.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Insertion sort.

2. Model answer

Decrease the exponent by the same number of places.

3. Model answer

¬A ∧ ¬B.

4. Model answer

0.75×8=6.

Vocabulary — know these terms

sign extension — Copy the sign bit into a new high-order position.

De Morgan — Swap AND/OR while negating components.

normalisation — Remove redundant leading mantissa sign bits.

mantissa — The fractional significant part.

exponent — Power-of-two scale.

NEW LEARNING

Read, annotate and explain

Show positive first

Use exactly eight bits for +11, rather than padding after converting.

Invert each bit

00001011 becomes 11110100.

Add one

11110100 + 1 = 11110101 for −11.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Model −6 in eight-bit two's complement.

Predict first

Write +6 as 00000110.

Check the method

Invert to 11111001.

Check the method

Add one to get 11111010, representing −6.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

(a) Represent −19 as an 8-bit two's-complement number. [4] (b) Perform an arithmetic right shift by 1 of your result. State its signed decimal interpretation. [4]

Question 2

(a) Use De Morgan's law to simplify ¬(A ∧ ¬B). [3] (b) Explain how an eight-cell group of ones simplifies a three-variable Karnaugh map. [5]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A positive two's-complement fractional mantissa is 0.0011_2 with exponent 5. Normalise it, explaining each shift and why the value is unchanged.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An arithmetic right shift of −19 by one place produces −9.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.